

Question 1 [30 marks]

Rectangular Snip

Lets assume that an array A contains 5 integer elements.

1. Read the elements of this array via console from the user. [6 marks]
2. Find the index of a specified element 'x' in the array A and display its index. If the element is not found in A, print -1. If multiple elements with the same value are present, print the first index. [8 marks]
 - E.g., A = [4, -1, 1, 10, 1],
 - x = 1, index = 2
 - x = 100, index = -1
3. Print the maximum value of the elements in A. [8 marks]
 - E.g., A = [4, -1, 1, 10, 1], maximum value = 10
4. Print the sum of the values of the elements in A. [8 marks]
 - E.g., A = [4, -1, 1, 10, 1], sum = 15

Question 2 [70 marks]

Write a class called "counter" which contains the following:

1. A public member variable "name" which represents the name of the counter. [5 marks]
2. A private member variable "value" which is an integer number, represents the value of the counter. [5 marks]
3. A constructor to initialize the name, and set the value of the counter to 0. [10 marks]
4. A public method "up" which will increase the value of the counter by 1. [10 marks]
5. A public method "down" which will decrease the value of the counter by 1. Note that the value of the counter will never be negative. [10 marks]
6. Setter and getter method to set and get the value of the counter. [10 marks]
7. A method called "print" to print the name and the current value of the counter. [10 marks]
8. Create a main method and create an instance of the "counter" class as follows. [10 marks]

8. Create a main method and create an instance of the "counter" class as follows. [10 marks]

```
counter c = new counter("Red");  
c.print();  
c.up();  
c.down();  
c.down();  
c.print();  
c.setValue(10);  
c.down();  
int tmp = c.getValue();  
System.out.println(tmp);
```

The output should be similar to the following.

```
Name of the counter: Red, value = 0  
Increasing, value = 1  
Decreasing, value = 0  
Decreasing, Operation is not permitted. value = 0  
Name of the counter: Red, value = 0  
Decreasing, value = 9
```